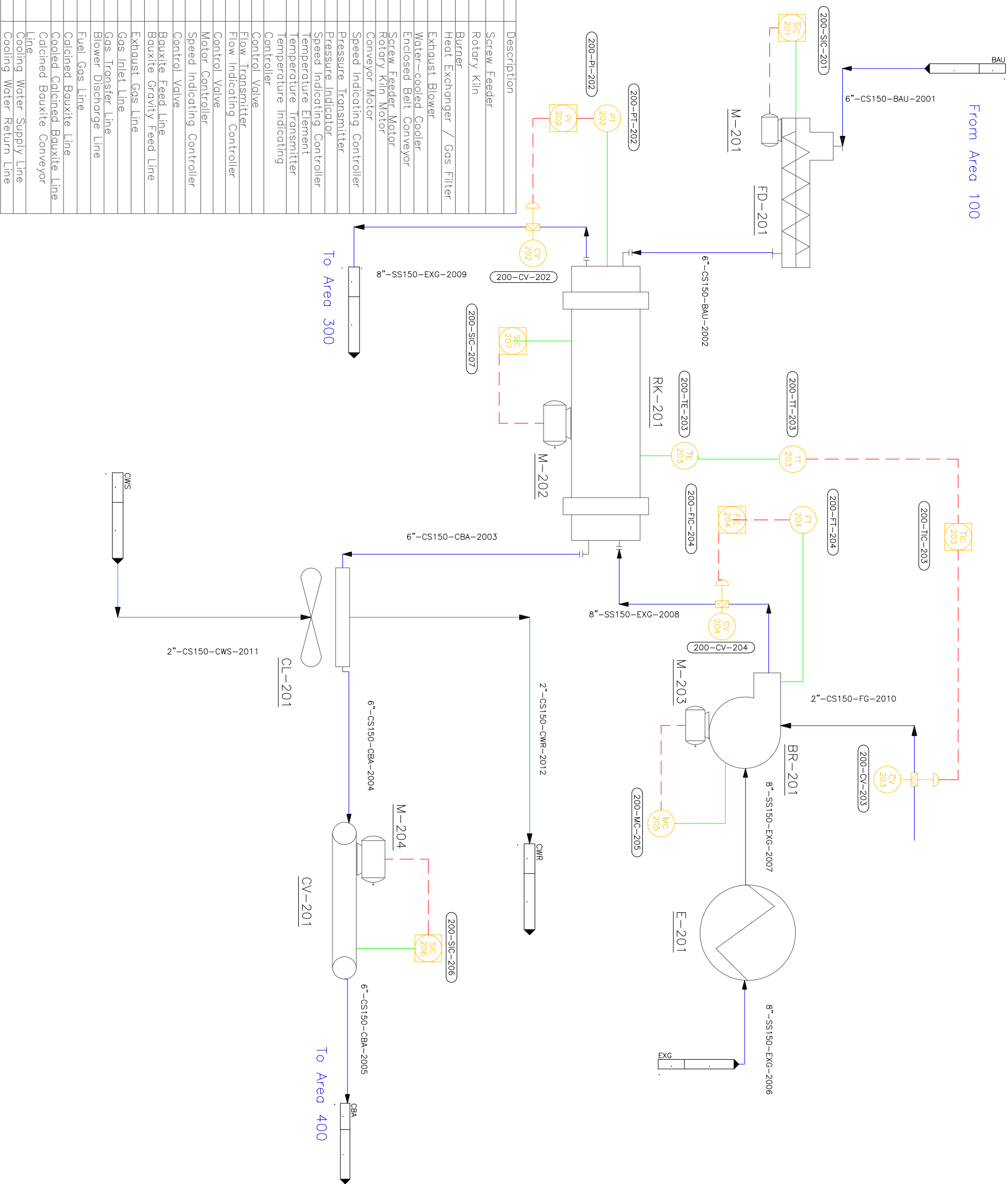


1. All pipes, fittings, and valves shall be designed in accordance with ASME B31.3 process piping standards.
2. All instrumentation symbols and nomenclature conform to ISA 5.1 standards.
3. Fail-safe actions for all pneumatic control valves
4. (fail-open/fail-closed) shall be determined during the HAZOP review phase.
5. Piping directly connected to the rotary kiln (RK-201) must be equipped with expansion joint to accommodate thermal growth.
6. Personnel protection insulation (PP) is required on all exposed surfaces and piping exceeding 60 degrees Celsius.
7. Kiln refractory lining specifications and thickness to be determined by maximum calcination temperature of bauxite.

No.	Revision/Issue	Date

Area200
Pyro-Processing
(Calcination &
Reduction)

Project Name and Address	
PAC_P&ID	
Final Year Project	
PMB, UPO, KNUST, Ghana	
Kumasi Ashanti Region	
Area	Stamp
200	
Drawing Number	
2	
Author	
GROUP 7B	



Tag Number	Category	Description
FD-201	Equipment	Screw Feeder
BR-201	Equipment	Rotary Kiln
E-201	Equipment	Burner
M-203	Equipment	Heat Exchanger / Gas Filter
CI-201	Equipment	Exhaust Blower
CV-201	Equipment	Water-cooled Cooler
M-201	Electrical	Enclosed Belt Conveyor
M-202	Electrical	Screw Feeder Motor
M-204	Electrical	Rotary Kiln Motor
SIC-201	Instrumentation	Conveyor Motor
PI-202	Instrumentation	Speed Indicating Controller
PI-202	Instrumentation	Pressure Transmitter
SIC-207	Instrumentation	Pressure Indicator
TE-203	Instrumentation	Speed Indicating Controller
TI-203	Instrumentation	Temperature Element
TIC-203	Instrumentation	Temperature Transmitter
CV-203	Instrumentation	Temperature Indicating Controller
FI-204	Instrumentation	Control Valve
FI-204	Instrumentation	Flow Transmitter
FI-204	Instrumentation	Flow Indicating Controller
CV-204	Instrumentation	Control Valve
MC-205	Instrumentation	Motor Controller
SIC-206	Instrumentation	Speed Indicating Controller
CV-202	Instrumentation	Control Valve
6"-CSI50-BAU-2001	Piping	Bauxite Feed Line
6"-CSI50-BAU-2002	Piping	Bauxite Gravity Feed Line
8"-SSI50-EXG-2009	Piping	Exhaust Gas Line
8"-SSI50-EXG-2006	Piping	Gas Inlet Line
8"-SSI50-EXG-2007	Piping	Gas Transfer Line
8"-SSI50-EXG-2008	Piping	Blower Discharge Line
2"-CSI50-FG-2010	Piping	Fuel Gas Line
6"-CSI50-CBA-2003	Piping	Calcined Bauxite Line
6"-CSI50-CBA-2004	Piping	Cooled Calcined Bauxite Line
6"-CSI50-CBA-2005	Piping	Calcined Bauxite Conveyor
2"-CSI50-CWR-2011	Piping	Line
2"-CSI50-CWR-2012	Piping	Cooling Water Supply Line
	Piping	Cooling Water Return Line